L2.1 Introduction to Machine Learning Quiz ANS

1. In supervised learning, what information does the teacher provide?
A) Example inputs and correct outputs
B) Only unlabeled input data
C) Rewards without desired outputs
D) A fixed policy with no examples

ANS: A) Example inputs and correct outputs

2. What is the primary goal of unsupervised learning?
A) To predict only continuous values
B) To discover structure or hidden patterns in unlabeled data
C) To maximize cumulative rewards through interaction
D) To compare predictions with teacher-provided labels

ANS: B) To discover structure or hidden patterns in unlabeled data

3. In reinforcement learning, what does an agent attempt to maximize?
A) The number of input features
B) The size of each mini-batch
C) Its cumulative rewards
D) The number of hidden layers

ANS: C) Its cumulative rewards

4. Which task is an example of regression?
A) Classifying an email as spam or not spam
B) Identifying an image as a cat or dog
C) Predicting whether a statement is true or false
D) Predicting a house price for next year

ANS: D) Predicting a house price for next year

5. Which statement correctly contrasts training and inference?
A) Training adjusts model parameters through repeated forward and backward passes, whereas inference typically uses one forward pass.
B) Training uses one forward pass, whereas inference requires millions of back-propagation iterations.
C) Training always runs on an edge device, whereas inference requires a GPU cluster.
D) Training and inference both update model weights in the same way.

ANS: A) Training adjusts model parameters through repeated forward and backward passes, whereas inference typically uses one forward pass.

6. Which type of value is normally predicted by a classification model?
A) A continuous salary value
B) A discrete class label
C) A model's training time
D) A learning-rate schedule

ANS: B) A discrete class label

7. What does a loss function measure in supervised learning?
A) The number of layers in a neural network
B) The size of the training dataset
C) How much the predicted value differs from the ground-truth value
D) The frequency of an edge processor

ANS: C) How much the predicted value differs from the ground-truth value

8. Which function is used in the lecture's logistic-regression model for binary classification?
A) Mean absolute error
B) K-fold averaging
C) Random cropping
D) Sigmoid function

ANS: D) Sigmoid function

9. What layers make up the two-layer fully connected neural network used to solve XOR in the presentation?
A) One input layer, one hidden layer, and one output layer
B) Two input layers and no output layer
C) One convolution layer and one pooling layer only
D) Five hidden layers and one input layer

ANS: A) One input layer, one hidden layer, and one output layer

10. What may happen if a neural network has too many layers and neurons?
A) It must become a regression model.
B) It may overfit the training data.
C) Its model capacity always decreases.
D) It can no longer use activation functions.

ANS: B) It may overfit the training data.

11. Which loss function is commonly associated with classification in the presentation?
A) Mean Squared Error only
B) Mean Absolute Error only
C) Cross-Entropy Loss
D) Quantile Loss only

ANS: C) Cross-Entropy Loss

12. Which loss function is commonly associated with regression?
A) Focal Loss
B) Hinge Loss
C) Cross-Entropy Loss
D) Mean Squared Error

ANS: D) Mean Squared Error

13. In multi-class image classification, what are the two main processing stages?
A) Feature extraction followed by classification
B) Back-propagation followed by data labeling
C) Map construction followed by vehicle control
D) Regression followed by reinforcement learning

ANS: A) Feature extraction followed by classification

14. What does the SoftMax output represent in a multi-class classifier?
A) A list of gradients whose values sum to zero
B) A list of class probabilities that sum to 1.0
C) A list containing only ground-truth labels
D) A list of hidden-layer sizes

ANS: B) A list of class probabilities that sum to 1.0

15. In a medical-test example, which outcome is a false positive?
A) A sick person is diagnosed as sick.
B) A healthy person is diagnosed as healthy.
C) A healthy person is misdiagnosed as sick.
D) A sick person is misdiagnosed as healthy.

ANS: C) A healthy person is misdiagnosed as sick.

16. In a medical-test example, which outcome is a false negative?
A) A healthy person is correctly diagnosed as healthy.
B) A healthy person is misdiagnosed as sick.
C) A sick person is correctly diagnosed as sick.
D) A sick person is misdiagnosed as healthy.

ANS: D) A sick person is misdiagnosed as healthy.

17. What is the purpose of back-propagation during neural-network training?
A) To calculate gradients used to update model parameters
B) To convert a regression problem into clustering
C) To replace the forward pass during inference
D) To divide data into cross-validation folds

ANS: A) To calculate gradients used to update model parameters

18. How does mini-batch stochastic gradient descent compute a parameter update?
A) It always uses the entire dataset for every gradient.
B) It uses a small portion of the training data to compute the gradient.
C) It uses validation data as ground-truth test data.
D) It updates parameters without calculating a loss.

ANS: B) It uses a small portion of the training data to compute the gradient.

19. What does a large gap between training accuracy and validation accuracy often imply?
A) The model has no trainable parameters.
B) The model is necessarily underfitting.
C) The model may be overfitting.
D) The validation set has become the training set.

ANS: C) The model may be overfitting.

20. In 5-fold cross-validation, how is each fold used?
A) All five folds are used only for testing.
B) One fold is permanently discarded.
C) Each fold is used only for training and never for validation.
D) Each fold serves as validation once while the other four serve as training data.

ANS: D) Each fold serves as validation once while the other four serve as training data.